

# Dongwook Kwon

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## SUMMARY

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Highly accomplished AI Research Engineer specializing in Large Language Model (LLM) optimization, Multi-Agent Systems (MAS), and AI-driven anomaly detection for critical infrastructure. Demonstrated research excellence through co-first authored publications in top-tier peer-reviewed venues, including an Oral presentation at EMNLP 2025 and a paper at ACL 2026 (both Industry Tracks), developed in collaboration with LG Electronics. Inventor of multiple registered patents in network and medical device anomaly detection. A core engineer who engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company. With foundational research experience in nuclear cybersecurity and medical cyber-physical systems, poised to significantly advance the United States' technological edge in critical and emerging fields by engineering scalable, reliable, and secure autonomous AI architectures.

## PATENTS

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### REGISTERED

Apr 2026 **Method for Implantable Medical Device to Detect Anomaly in Real Time**, KR 10-2961984

Aug 2025 **GPT API-Based Network Anomaly Detection**, KR 10-2848814

### PENDING APPLICATIONS

Oct 2025 **System for Anomaly Detection Using a Collaborative Multi-AI Determination Module Based on LLM**, KR 10-2025-0160217

Aug 2024 **Categorical Data of Networks Based Network Anomaly Detection Apparatus and Method Thereof**, KR 10-2024-0110559

## PUBLICATIONS

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### JOURNAL ARTICLES

Y. Nam, J. Lee, H. Lee, **D. Kwon**, M. Yeo, S. Kim, R. Sohn, and C. Park, "Designing a remote photoplethysmography-based heart rate estimation algorithm during a treadmill exercise," *Electronics*, vol. 14, no. 5, p. 890, 2025.

**D. Kwon**, Y. Kang, J. Lee, Y. Nam, J. Won, K. K. Kim, D. D. Kim, and C. Park, "Graph-Enhanced Bidirectional GRU for Anomaly Detection in Power Generation Environments," under review at *Data Mining and Knowledge Discovery (Springer Nature)*, 2025.

### CONFERENCE PAPERS

R. Chang<sup>†</sup>, **D. Kwon**<sup>†</sup>, J. Lee<sup>†</sup>, and N. Verma, "CascadeDebate: Multi-Agent Deliberation for Cost-Aware LLM Cascades," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguistics (ACL) — Industry Track (Poster)*, 2026. <sup>†</sup>Equal contribution.

J. Lee<sup>†</sup>, R. Chang<sup>†</sup>, **D. Kwon**<sup>†</sup>, H. Singh, and N. Verma, "GEMMAS: Graph-based Evaluation Metrics for Multi Agent Systems," in *Proceedings of the EMNLP 2025 Industry Track (Oral)*, 2025. <sup>†</sup>Equal contribution.

**D. Kwon**, M. Kim, Y. Kang, J. Lee, and C. Park, "Hybrid neural network model for anomaly detection in implantable devices using graph attention networks and transformers," in *Proc. 10th Int. Biomed. Eng. Conf. (IBEC)*, 2024. **Best Poster Award — Silver.**

**D. Kwon**, Y. Kang, and C. Park, "Enhancing medical device security with GNN-GRU anomaly detection model," in *Proc. 62nd KOSOMBE Autumn Conf.*, pp. 204–205, 2023. **Outstanding Paper Award.**

J. Yang, J. Kim, J. Lee, H. Ryu, **D. Kwon**, S. Yeo, P. Kim, Y. Kim, J. Lim, H. Yoon, and C. Park, "Metaverse: Design of the Car Price Prediction Model Through a Machine-learning Approach," in *Proc. IEEE Int. Conf. Metaverse Computing, Networking and Applications (MetaCom)*, pp. 734–737, 2023.

RESEARCH & PROFESSIONAL EXPERIENCE

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| <b>AI Engineer, Intelligent Agent Technology Team</b><br><i>Suresoft Technologies</i>                                                                                                                                                                                                                                                                                                                                                  | Dec. 2025 – Present<br><i>Seongnam, Korea</i>       |
| <ul style="list-style-type: none"><li>• Developing an AI validation system utilizing AI agent-based architectures for enterprise model verification.</li><li>• Designing autonomous and collaborative AI agents that verify and evaluate machine-learning models at scale.</li><li>• Engineered and delivered a critical agent-based codebase for a mission-critical vehicle control data project for Hyundai Motor Company.</li></ul> |                                                     |
| <b>Research Assistant</b><br><i>Bio-Computing &amp; ML Lab, Kwangwoon University</i>                                                                                                                                                                                                                                                                                                                                                   | Aug. 2021 – Dec. 2025<br><i>Seoul, Korea</i>        |
| <ul style="list-style-type: none"><li>• Researched deep-learning algorithms for anomaly detection in complex time-series data, focusing on critical cyber-physical systems including nuclear power generation environments.</li><li>• Co-developed an Active Kill-Switch and Biomarker-Based Defense System for life-threatening IoT medical devices.</li></ul>                                                                        |                                                     |
| <b>Visiting Researcher</b><br><i>LG Electronics Canada</i>                                                                                                                                                                                                                                                                                                                                                                             | Jan. 2025 – Jul. 2025<br><i>Toronto, ON, Canada</i> |
| <ul style="list-style-type: none"><li>• Built an LLM-driven assessment framework for evaluating performance of agentic AI systems.</li><li>• Work fed directly into peer-reviewed publications at EMNLP 2025 and ACL 2026.</li></ul>                                                                                                                                                                                                   |                                                     |
| <b>AI Development Project Participant, Summer Program</b><br><i>Qualcomm Institute, UC San Diego</i>                                                                                                                                                                                                                                                                                                                                   | Jul. 2022 – Aug. 2022<br><i>San Diego, CA, USA</i>  |
| <ul style="list-style-type: none"><li>• Built a personality-based drug-addiction prediction system using machine-learning models.</li><li>• Recognized with the program's Achievement Award.</li></ul>                                                                                                                                                                                                                                 |                                                     |

EDUCATION

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| <b>Kwangwoon University</b><br><i>M.S. in Computer Engineering, GPA 4.33 / 4.5 (98.1%)</i>                       | Seoul, Korea<br>Mar. 2024 – Feb. 2026        |
| <b>University of Toronto</b><br><i>Graduate Research Program, GPA 3.68 / 4.0</i>                                 | Toronto, ON, Canada<br>Jan. 2025 – Jul. 2025 |
| <b>Kwangwoon University</b><br><i>B.S. in Electronics and Communications Engineering, GPA 3.41 / 4.5 (88.1%)</i> | Seoul, Korea<br>Mar. 2019 – Feb. 2024        |

HONORS & AWARDS

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| MAJOR AWARDS                                                                                                               |  |
| Aug. 2022 <b>Achievement Award</b> , Qualcomm Institute AI Development Project — UC San Diego                              |  |
| Dec. 2021 <b>Gold Award (Ministerial Award)</b> , 2021 Smart Maritime Logistics Project — Minister of Oceans and Fisheries |  |
| ACADEMIC RECOGNITION                                                                                                       |  |
| Nov. 2025 <b>Outstanding Student Paper Award</b> , Conference of the Institute of Electronics and Information Engineers    |  |
| Nov. 2024 <b>Best Poster Award (Silver)</b> , 10th International Biomedical Engineering Conference (IBEC 2024)             |  |
| Nov. 2023 <b>Outstanding Paper Award</b> , Conference of the Korean Society of Medical and Biological Engineering          |  |